

the creation basket (which is cheap). As they sell, the price of the ETF declines until it reaches the NAV.

How close is the NAV of the shares, at which the authorized participants trade, and the market price of the ETF, at which other investors trade? Pretty close, most of the time. Engle and Sarkar (2006) and DeFusco, Ivanov, and Karels (2011) show that ETFs are efficiently priced. The deviations from NAV are small, not persistent, and last only minutes.

Flash Crash

Except sometimes deviations are large, persistent, and last for more than just minutes.

In a mere 20 minutes starting at 2:40 pm on May 6, 2010, major futures and equities markets plummeted more than 5% before suddenly rebounding. The decline began in the equity futures market, where it was (supposedly) triggered by Waddell & Reed, an investment manager based in Kansas. Pressure from high-frequency trading algorithms soon overwhelmed available liquidity, causing an explosion of mispricing to tumble through to equities markets.⁵⁵ This was the *Flash Crash*. More than 300 securities had trades executed at prices more than 60% away from values just before the Flash Crash. Accenture, a company that has a market value in the tens of billions of dollars, traded at 1 cent. Sotheby's had been trading around \$34 and then jumped to \$99,999.99. This gave Sotheby's, however briefly, a market capitalization larger than the U.S. economy.

ETFs accounted for 70% of the 326 securities for which trades were canceled. Madhavan (2012) shows that ETFs experienced significantly larger drawdowns during the Flash Crash than for other securities: the average ETF drawdown was 24% versus 8% for other equities. ETFs played a special role in transmitting these shocks. Ben-David, Franzoni, and Moussawi (2011) argue that ETFs served as the conduit for shocks from the futures market (ground zero) to the equity market. It is not surprising that some regulators and others think that ETFs pose systemic risk to the financial system.⁵⁶

There are several lessons for asset owners from the Flash Crash:

1. ETFs require liquid markets for fair pricing;
2. ETFs can exacerbate liquidity shocks; and
3. ETF no-arbitrage relationships are prone to breakdown and are very fragile during crashes.

⁵⁵ See U.S. Securities and Exchange Commission and the Commodity Futures Trading Commission, "Findings Regarding the Market Events of May 6, 2010," released on Sept. 30, 2010. Waddell & Reed was not named by the SEC and CFTC report. See Lash, H., and Spicer, J., "Exclusive: Waddell is Mystery Trader in Market Plunge," Reuters, May 15, 2010.

⁵⁶ See Bradley and Litan (2010).